Chapter 6

1.Describe RAID working mechanism.

RAID works by placing data on multiple disks and allowing input/output (I/O) operations to overlap in a balanced way, improving performance. Because using multiple disks increases the mean time between failures, storing data redundantly also increases fault tolerance.

RAID arrays appear to the operating system (OS) as a single logical drive. RAID employs the techniques of disk mirroring or disk striping. Mirroring will copy identical data onto more than one drive. Striping partitions help spread data over multiple disk drives. Each drive's storage space is divided into units ranging from a sector of 512 bytes up to several megabytes. The stripes of all the disks are interleaved and addressed in order. Disk mirroring and disk striping can also be combined in a RAID array.

Chapter 5

Key properties of semiconductor memory:

✓ Stores data in magnetic form.

✓ Affected by magnetic fields.

✓ Has high storage capacity.

✓ Doesn't use a laser to read/write data.

✓ Magnetic storage devices are; Hard disk, Floppy disk, Magnetic tape etc.

Define memory cell, Describe the structure of a typical memory cell.

▪ A memory cell is a small unit of storage in a computer's memory, typically used to store a single byte of data.

▪ Today, the most common memory cell architecture is MOS memory, which consists of metal–oxide–semiconductor (MOS) memory cells. Modern random access memory (RAM) uses MOS field-effect transistors (MOSFETs) as flip-flops, along with MOS capacitors for certain types of RAM.

b) How error-correcting code works? Describe the Hamming Error-correcting code procedure.

- **An error correcting code (ECC)** is an encoding scheme that transmits messages as binary numbers, in such a way that the message can be recovered even if some bits are erroneously flipped. They are used in practically all cases of message transmission, especially in data storage where ECCs defend against data corruption.
- **Hamming error-correcting code procedure:**
  - ✓ Add extra "parity" bits to your data.
  - ✓ Position parity bits at powers of 2 (1, 2, 4, 8, etc.).
  - ✓ Calculate parity bit values based on data bits.
  - ✓ Transmit data with parity bits.
  - ✓ Receiver checks parity bits for errors.
  - ✓ If errors detected, correct them using parity bits.
  - ✓ If multiple errors occur, detect but can't correct.

3. How can you characterize the memory of your computer?

In general, memory can be divided into primary and secondary memory; moreover, there are numerous types of memory when discussing just primary memory. Some types of primary memory include the following

- **Cache memory:** This temporary storage area, known as a cache, is more readily available to the processor than the computer's main memory source.
- **RAM:** The term is based on the fact that any storage location can be accessed directly by the processor.
- **Dynamic RAM:** DRAM is a type of semiconductor memory that is typically used by the data or program code needed by a computer processor to function.
- **Static RAM:** SRAM retains data bits in its memory for as long as power is supplied to it. Unlike DRAM, which stores bits in cells consisting of a capacitor and a transistor, SRAM does not have to be periodically refreshed.
- **Double Data Rate SDRAM:** DDR SRAM is SDRAM that can theoretically improve memory clock speed to at least 200 MHz.